5. Learned Models and Online Adaptation

Chapter 4 established how the datasets were generated and why both closed-loop operation and open-loop valve excitation were required. This chapter develops the learned representations that consume them: the inverse-dynamics (ID) and forward-dynamics (FD) networks of the IF actuator path, the command-conditioned reduced-order model (ROM) that replaces them in R, and the system-dynamics surrogate, privileged factor encoder and Arm Adaptation Module (AAM) that together carry the adaptation argument.

Two commitments organise the chapter. The first is that the surrogate is kept strictly separate from the deployed actuator: the system-dynamics model is an instrument for establishing whether a compact dynamics latent exists and is useful, never a component of any control stack. The second is that no model is accepted on a supervised score because a recurring result of this work is that such scores can be not merely optimistic but uninformative: a forward model reached a one-step R^2 near 0.999 while having learned almost nothing about valve authority (Section 5.3). Every model is therefore presented with its contract, training objective, and the offline evidence that accepted or refused it. All models operate on cylinder-space signals; the mapping to task space belongs to Chapter 3, the control integration to Chapter 6, and deployment-level results to Chapter 7.

5.1 Modelling Overview and Common Notation

All learned models act on the four hydraulic cylinders of the MULAG arm, indexed $i \in \{1, \dots, 4\}$ in the fixed order slew cylinder (*Drehzapfen*), boom cylinder I (*Ausleger I*), boom cylinder II (*Ausleger II*) and head pivot cylinder (*Messerkopf-Schwenk*). The passive fifth joint is not actuated and enters no model input. At the learned-model sample time $T = 0.05\,\mathrm{s}$ the discrete index k refers to the cylinder extension $\boldsymbol{x}_k$, the extension velocity $\dot{\boldsymbol{x}}_k$, the chamber pressure difference $\Delta\boldsymbol{p}_k$ and the logged rod force $\boldsymbol{F}_k$, each in $\mathbb{R}^4$. The valve command $\boldsymbol{u}_k \in [-1, 1]^4$ is the normalised spool command of the four proportional valves. The desired cylinder command produced by the task controller is the pair

$$\boldsymbol{c}_k = \left[\Delta\boldsymbol{x}_k^*,\ \Delta\dot{\boldsymbol{x}}_k^*\right] \in \mathbb{R}^8, \tag{5.1}$$